Recheck - [Q: 8] [Part B - 2, 18]
Part A

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
2023-2024, II Semester, Dated 30/11/2023
CE F230 (Civil Engineering Materials) Lab Quiz

24/40

27

Max. Marks: 40 **Duration: 30 minutes**
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PART A: Choose the best answer: (1 M)

1. The correct expression for calculating the target mean strength is $f_{ck}' = f_{ck} + k \times S$? The value of "k" is
a)1.65 b)1.56 c)1.50 d)1.20 X

2. As per IS 3495-1992, the depth of immersion of bricks to find efflorescence is___?
a)15mm b) 25mm c) 35mm d) 45 mm 1

3. The function of fine aggregate is?
a) To assist in producing workability and uniformity in the mixture
b) To assist the cement paste to hold the course aggregate particles in suspension
c) To promote the plasticity in the mixture and prevent possible segregation of paste and coarse aggregates
d) All of the above 1

4. Bulking of sand is caused due to:
a) Air voids b) Viscosity c) Clay content d) Surface moisture 1

5. Silt content in fine aggregate causes:
a) Gain in early strength due to its fineness
b) Loss in strength as it hampers hydration and bonding between cement-aggregate particles 1
c) Plays no role
d) None of the above

6. Calorimeter apparatus used to measure heat of hydration of cement. 1

7. Concrete made from aggregate grading having least surface area will require least water, and will consequently be the strongest
a) True b) False X

8. Choose the correct statement:
a) Water is added at the rate of P/4 + 3% for Tensile strength test of cement.
b) Water is added at the rate of P/5 + 4% for Tensile strength test of cement.
c) Water is added at the rate of P/5 + 3% for Tensile strength test of cement. X
d) None of the above

9. The principal on which rebound hammer works:
a) Compressibility b) Ductility c) Surface hardness d) None 1

10. Efflorescence in brick is due to deposition of soluble salts of
a. Potassium
b. Calcium 1
c. Sodium
d. All of the above

11. Method of loading in split tensile test:
a. Line load parallel to cylinder axis
b. Area load perpendicular to cylinder axis X
c. Line load perpendicular to cylinder axis
d. Area load parallel to cylinder axis

PART B: Answer the following/ Choose best answer: (1 M)

1. Name the apparatus: (1*5=5) 5

Sieve	Ultrasonic wave emitter	travel	vicket	slump cone

2. For Compressive Strength, if the water content required for cement paste is 360 mL, the weight of the dry sample will be 3600 gm ? (Consider normal consistency as 28). X

3. Load of 50kN is applied in split tensile experiment and dimensions of cylinder sample as follows: diameter - 12cm, Length = 25cm. Then the split tensile strength is 1.06 N/mm² 1

4. The standard size of the red bricks is 25cm X 10cm X 1.5 cm X

5. To determine the fineness of cement, standard sieve of size 75 μ is used. X

$$\left(\frac{28}{4}+3\right) \times x = 3600$$
$$10x = 3600$$

$$T = \frac{2 \times 50 \times 1000}{\pi \times 250 \times 120}$$

W_1, W_2, W_3, W_4

6. An experiment was conducted to calculate the specific gravity of sand using a pycnometer. The specific gravity was calculated as 2.32. The weight of water with pycnometer was 1550 g. Weight of pycnometer with sand and water was 1750 g. Calculate the amount of sand used in the experiment is 200 g   1

7. The flexural tensile strength of a concrete beam specimen with breaking load 10 kN, span length 50 ~~mm~~ cm, depth and width 10 cm and 5 cm respectively is ________

8. Flexure strength is determined as:
a) Modulus of rigidity b) Modulus of rupture c) Modulus of plasticity d) Modulus of elasticity   X

$$2.32 = \frac{W_2 - W_1}{(W_4 - W_1) - (W_3 - W_2)}$$

9. Aggregate is called fine aggregate if it is passes through which IS sieve?
a) 2.36mm b) 45μ c) 4.75mm d) 75μ   1

10. The specific gravity of sand is around 2.56   1

11. To determine the flexural strength, the size of test specimen used is:

a) 150 x150 x500 mm b) 100 x100 x700 mm c) 150 x150 x700 mm d) 100 x100 x500 mm   X

12. Standard consistency of cement is tested by vicket apparatus.   1
vicat

13. The figure below represents a:

a) Low slump b) Normal slump c) Shear slump d) Collapse slump   1

14. Write down the full form of C-S-H gel ________________

15. How is bulking related to moisture content?
a) Keeps on increasing with moisture content b) Increases till a certain point and then decreases   1
c) Keeps on decreasing with moisture content d) Decreases till a certain point and then increases

16. Raw materials for fly ash bricks are:
I. Fly ash; II Lime; III Clay; IV Gypsum; V sand; VI Coarse aggregates.
a) I, II, V, VI b) I, III, IV, V c) I, II, IV, V d) ~~I, III, IV, V~~ None of there   1

17. IS code for the efflorescence test of clay bricks is
a) IS 3495 Part 3 b) IS 3495 Part 2 c) IS 3495 Part 4 d) IS 3495 Part 1   X

18. What is the purpose of the pan at the bottom of the stack of sieves during a sieve analysis?
a) To catch the material passing through the finest sieve b) To support the entire stack of sieves   X
c) To prevent the loss of fines during shaking d) To measure the density of the sample

19. What property of concrete is primarily assessed by Ultrasonic Pulse Velocity Testing?
a) Compressive Strength b) Tensile Strength c) Permeability d) Density   X

20. The RCPT is primarily used to assess the
a) Compressive strength of concrete b) Chloride ion penetration resistance of concrete   X
c) Concrete's resistance to carbonation d) Surface hardness of concrete

21. The cement mortar briquettes are used to determined:
a) Compression strength b) Tensile strength c) Impact strength d) Shear strength   1

22. Which shape of concrete specimen displays higher compressive strength for same parameters, cube or cylinder   1
Cube

23. In a stress-strain curve for tensile test of steel rebar, the point where plastic deformation begins is known as: X
a) Elastic limit b) Yield point c) Ultimate tensile strength d) Breaking point

24. What is the typical shape of a stress-strain curve for steel rebars till elastic limit during a tensile test?   1
a) Linear b) Hyperbolic c) Sigmoidal d) Parabolic

applied force   Cross Sectional area
25. The ratio of the ~~cross-sectional area~~ of the specimen to the ~~applied force~~ is known as: 1
a) Strain b) Stress c) Modulus of elasticity d) Ductility